

ERRATUM: GLOBAL REGULARITY OF NONDIFFUSIVE TEMPERATURE FRONTS FOR THE 2D VISCOUS BOUSSINESQ SYSTEM*

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Abstract. There is an error of misusing a commutator formula in the proof of Lemma 2.7 in [SIAM J. Math. Anal., 54 (2022), pp. 4043–4103] so that this lemma is not correct. However, by establishing a weaker version of this commutator lemma, the main results including Theorem 1.1 in [SIAM J. Math. Anal., 54 (2022), pp. 4043–4103] still hold true. In addition, we confirm the validity of Lemmas 2.4(3) and 2.5(3) in [SIAM J. Math. Anal., 54 (2022), pp. 4043–4103] although a similar error occurs in the proof.

Key words. Boussinesq system, temperature patch problem, commutator estimate, striated estimates

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In [1], there is an error of misusing a commutator formula in the proof of Lemma 2.7 so that the commutator estimate (2.23) is not valid. But we instead can show a weaker result as follows, which is sufficient for our use.

LEMMA 0.1. *Let $s \in (0, 1)$, $p \in [2, \infty]$, $r \in [1, \infty]$. Let $\mathcal{R}_{-1} := m(D)\Lambda^{-1}$, $\Lambda = (-\Delta)^{1/2}$, and let $m(D)$ be a zero-order pseudo-differential operator with its symbol $m(\xi) \in C^\infty(\mathbb{R}^d \setminus \{0\})$. Assume that $u = (u_1, \dots, u_d)$ is a smooth divergence-free vector field and ϕ is a smooth scalar function. Then we have*

$$(0.1) \quad \|[\mathcal{R}_{-1}, u \cdot \nabla] \phi\|_{B_{p,r}^s} \leq C \|\nabla u\|_{L^p} (\|\phi\|_{B_{\infty,r}^{s-1}} + \|\phi\|_{L^2}) + C \|u\|_{L^2} \|\phi\|_{L^2},$$

where $C > 0$ is a constant depending only on s , p , and d .

Proof. Bony's decomposition gives

$$\begin{aligned} & [\mathcal{R}_{-1}, u \cdot \nabla] \phi \\ &= \sum_{q \in \mathbb{N}} [\mathcal{R}_{-1}, S_{q-1} u \cdot \nabla] \Delta_q \phi + \sum_{q \in \mathbb{N}} [\mathcal{R}_{-1}, \Delta_q u \cdot \nabla] S_{q-1} \phi + \sum_{q \geq -1} [\mathcal{R}_{-1}, \Delta_q u \cdot \nabla] \tilde{\Delta}_q \phi \\ &=: \text{I} + \text{II} + \text{III}, \end{aligned}$$

where we have adopted the standard notations in the Littlewood–Paley theory (see [1, section 2]). For I, there exists a bump function $\tilde{\psi} \in C_c^\infty(\mathbb{R}^d)$ supported on an annulus of $\mathbb{R}^d$ away from zero such that

$$(0.2) \quad \text{I} = \sum_{q \in \mathbb{N}} [\mathcal{R}_{-1} \tilde{\psi}(2^{-q} D), S_{q-1} u \cdot \nabla] \Delta_q \phi$$

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with $\mathcal{R}_{-1}\tilde{\psi}(2^{-q}D) = 2^{q(d-1)}\bar{h}(2^q\cdot)^*$ and $\bar{h} \in \mathcal{S}(\mathbb{R}^d)$. Thus by using Minkowski's inequality we find that for every $j \geq -1$,

$$\begin{aligned} 2^{js}\|\Delta_j \mathbf{I}\|_{L^p} &\leq C2^{js} \sum_{q \in \mathbb{N}, |q-j| \leq 4} \|[\mathcal{R}_{-1}\tilde{\psi}(2^{-q}D), S_{q-1}u \cdot \nabla] \Delta_q \phi\|_{L^p} \\ &\leq C2^{js} \sum_{|q-j| \leq 4} \int_{\mathbb{R}^d} 2^{q(d-1)} |\bar{h}(2^q y)| |y| dy \|\nabla S_{q-1}u\|_{L^p} \|\nabla \Delta_q \phi\|_{L^\infty} \\ &\leq C\|\nabla u\|_{L^p} \sum_{|q-j| \leq 4} 2^{q(s-1)} \|\Delta_q \phi\|_{L^\infty}, \end{aligned}$$

which ensures that

$$\|\mathbf{I}\|_{B_{p,r}^s} \leq C\|\nabla u\|_{L^p} \|\phi\|_{B_{\infty,r}^{s-1}}.$$

The error in [1, Lemma 2.7] appears in the treating of II, where it was thought a similar formula (0.2) holds for II, but indeed it is not correct. Instead, for II, noting that

$$\text{II} = \sum_{q \in \mathbb{N}} \left(\mathcal{R}_{-1}\tilde{\psi}(2^{-q}D) (\Delta_q u \cdot \nabla S_{q-1}\phi) - \Delta_q u \cdot \nabla \mathcal{R}_{-1}S_{q-1}\phi \right),$$

and using the fact $\mathcal{R}_{-1}\nabla \Delta_j$ ($j \in \mathbb{N}$) is bounded on $L^p(\mathbb{R}^d)$ for $p \in [2, \infty]$, we obtain

$$\begin{aligned} 2^{js}\|\Delta_j \text{II}\|_{L^p} &\leq C2^{js} \sum_{q \in \mathbb{N}, |q-j| \leq 4} (2^{-q} \|(\Delta_q u \cdot \nabla S_{q-1}\phi)\|_{L^p} + \|\Delta_q u\|_{L^p} \|\nabla \mathcal{R}_{-1}S_{q-1}\phi\|_{L^\infty}) \\ &\leq C \sum_{q \in \mathbb{N}, |q-j| \leq 4} 2^{qs} \|\Delta_q u\|_{L^p} \left(\|S_{q-1}\phi\|_{L^\infty} + \|\nabla \mathcal{R}_{-1}\Delta_{-1}\phi\|_{L^\infty} + \sum_{0 \leq l \leq q-1} \|\Delta_l \phi\|_{L^\infty} \right) \\ &\leq C\|\nabla u\|_{L^p} \sum_{|q-j| \leq 4} \left(2^{q(s-1)} \|\Delta_{-1}\phi\|_{L^2} + \sum_{0 \leq l \leq q-1} 2^{(q-l)(s-1)} 2^{l(s-1)} \|\Delta_l \phi\|_{L^\infty} \right), \end{aligned}$$

which combined with the discrete Young's inequality leads to that for every $s < 1$,

$$\|\text{II}\|_{B_{p,r}^s} \leq C\|\nabla u\|_{L^p} (\|\phi\|_{B_{\infty,r}^{s-1}} + \|\phi\|_{L^2}).$$

For III, in light of the divergence-free property of u , we split it as the following:

$$\begin{aligned} \text{III} &= \sum_{q \geq 3} \mathcal{R}_{-1}\nabla \cdot (\Delta_q u \tilde{\Delta}_q \phi) - \sum_{q \geq 3} \Delta_q u \cdot \nabla \mathcal{R}_{-1}\tilde{\Delta}_q \phi + \sum_{-1 \leq q \leq 2} [\mathcal{R}_{-1}\nabla \cdot, \Delta_q u] \tilde{\Delta}_q \phi \\ &=: \text{III}_1 + \text{III}_2 + \text{III}_3. \end{aligned}$$

For III_1 , since $\mathcal{R}_{-1}\nabla$ is bounded on $L^p(\mathbb{R}^d)$ for $p \in [2, \infty)$, we deduce that for $j = -1$ (realizing $\frac{2p}{p+2} = 2$ for $p = \infty$),

$$\begin{aligned} 2^{-s}\|\Delta_{-1}\text{III}_1\|_{L^\infty} &\leq C \sum_{q \geq 3} \|\Delta_q u \tilde{\Delta}_q \phi\|_{L^{\frac{2p}{p+2}}} \\ &\leq C \sum_{q \geq 3} \|\Delta_q u\|_{L^p} \|\tilde{\Delta}_q \phi\|_{L^2} \leq C\|\nabla u\|_{L^p} \|\phi\|_{L^2}, \end{aligned}$$

and for every $j \in \mathbb{N}$ and $s > 0$,

$$\begin{aligned} 2^{js} \|\Delta_j \text{III}_1\|_{L^p} &\leq C 2^{js} \sum_{q \geq 3, q \geq j-3} \|\Delta_j \mathcal{R}_{-1} \nabla \cdot (\Delta_q u \tilde{\Delta}_q \phi)\|_{L^p} \\ &\leq C \sum_{q \geq 3, q \geq j-3} 2^{(j-q)s} 2^q \|\Delta_q u\|_{L^p} 2^{q(s-1)} \|\tilde{\Delta}_q \phi\|_{L^\infty} \\ &\leq C c_j \|\nabla u\|_{L^p} \|\phi\|_{B_{\infty,r}^{s-1}}, \end{aligned}$$

where $\{c_j\}_{j \in \mathbb{N}}$ is such that $\|c_j\|_{\ell^r} = 1$. The estimation of III_2 is similar as that of III_1 , and we get

$$\|\text{III}_1\|_{B_{p,r}^s} + \|\text{III}_2\|_{B_{p,r}^s} \leq C \|\nabla u\|_{L^p} (\|\phi\|_{B_{\infty,r}^{s-1}} + \|\phi\|_{L^2}).$$

For III_3 , we directly have

$$\begin{aligned} \|\text{III}_3\|_{B_{p,r}^s} &\leq C \sum_{-1 \leq j \leq 6} \sum_{-1 \leq q \leq 2} \left(\|\Delta_j \mathcal{R}_{-1} \nabla \cdot (\Delta_q u \tilde{\Delta}_q \phi)\|_{L^p} + \|\Delta_q u \cdot \nabla \mathcal{R}_{-1} \tilde{\Delta}_q \phi\|_{L^p} \right) \\ &\leq C \sum_{-1 \leq q \leq 2} \left(\|\Delta_q u \tilde{\Delta}_q \phi\|_{L^1} + \|\Delta_q u \cdot \nabla \mathcal{R}_{-1} \tilde{\Delta}_q \phi\|_{L^1} \right) \\ &\leq C \sum_{-1 \leq q \leq 2} \|\Delta_q u\|_{L^2} \|\tilde{\Delta}_q \phi\|_{L^2} \leq C \|u\|_{L^2} \|\phi\|_{L^2}. \end{aligned}$$

Therefore, collecting the above estimates yields the wanted estimate (0.1). $\square$

We can use Lemma 0.1 to replace [1, Lemma 2.7] in the application, which is mainly used in section 3 and Appendix B of [1]. Indeed, in (3.10) and (3.20) of [1], one needs to estimate $\|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_T^1(L^2)}$ and $\|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_T^1(L^p)}$ ($p > 2$), and they both are controlled by the following:

$$\begin{aligned} &\|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_T^1(B_{2,1}^{1-\frac{2}{p}})} \\ &\leq C \|\nabla u\|_{L_T^1(L^2)} \left(\|\theta\|_{L_T^\infty(B_{\infty,1}^{-\frac{2}{p}})} + \|\theta\|_{L_T^\infty(L^2)} \right) + C \|u\|_{L_T^1(L^2)} \|\theta\|_{L_T^\infty(L^2)} \\ &\leq C \|u\|_{L_T^1(H^1)} \|\theta\|_{L_T^\infty(L^2 \cap L^\infty)}; \end{aligned}$$

while in (3.45) and (B.7), it suffices to control $\|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_t^1(B_{\infty,1}^{\gamma'})}$ and $\|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_t^1(B_{r,1}^{\gamma'})}$ for every $0 < \gamma' < 1 - \frac{2}{p}$ and $r \geq 2$, and from the Besov embedding they can be bounded as follows:

$$\begin{aligned} &\|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_t^1(B_{p,1}^{\gamma'+\frac{2}{p}})} + \|[\mathcal{R}_{-1}, u \cdot \nabla] \theta\|_{L_t^1(B_{2,1}^{\gamma'})} \\ &\leq C \|\nabla u\|_{L_t^1(L^p \cap L^2)} \left(\|\theta\|_{L_t^\infty(B_{\infty,1}^{\gamma'+\frac{2}{p}-1})} + \|\theta\|_{L_t^\infty(B_{\infty,1}^{\gamma'-1} \cap L^2)} \right) + \|u\|_{L_t^1(L^2)} \|\theta\|_{L_t^\infty(L^2)} \\ &\leq C \|\nabla u\|_{L_t^1(L^p \cap L^2)} \|\theta\|_{L_t^\infty(L^2 \cap L^\infty)} + \|u\|_{L_t^1(L^2)} \|\theta\|_{L_t^\infty(L^2)}. \end{aligned}$$

Hence, the main results including Theorem 1.1 in [1] still hold true.

Finally, we remark that the error of misusing commutator formula also appears in the proof of Lemmas 2.5-(3) and 2.4-(3) in [1], but they can be easily fixed without using the commutator structure, so the validity of Lemmas 2.5-(3) and 2.4-(3) is not

affected. Indeed, in Lemma 2.4-(3), it suffices to estimate $\|\mathcal{II}\|_{B_{p,r}^{-\epsilon}}$ with $\epsilon \in (-1, 1)$, $(p, r) \in [1, \infty]^2$, and

$$\mathcal{II} := \sum_{j \in \mathbb{N}} [m(D), \Delta_j u \cdot \nabla] S_{j-1} \phi = \mathcal{II}_1 - \mathcal{II}_2,$$

and

$$\mathcal{II}_1 := \sum_{j \in \mathbb{N}} m(D) \tilde{\psi}(2^{-j} D) (\Delta_j u \cdot \nabla S_{j-1} \phi), \quad \mathcal{II}_2 := \sum_{j \in \mathbb{N}} \Delta_j u \cdot \nabla m(D) S_{j-1} \phi,$$

where $m(D)$ is a zero-order pseudo-differential operator with symbol $m(\xi) \in C^\infty(\mathbb{R}^d \setminus \{0\})$ and $\tilde{\psi}$ is defined as in (0.2), then we have that for $q \geq -1$,

$$\begin{aligned} & 2^{-q\epsilon} \|\Delta_q \mathcal{II}\|_{L^p} \\ & \leq C 2^{-q\epsilon} \sum_{j \in \mathbb{N}, |j-q| \leq 4} \left(\|m(D) \tilde{\psi}(2^{-j} D) (\Delta_j u \cdot \nabla S_{j-1} \phi)\|_{L^p} + \|\Delta_j u \cdot \nabla m(D) S_{j-1} \phi\|_{L^p} \right) \\ & \leq C 2^{-q\epsilon} \sum_{j \in \mathbb{N}, |j-q| \leq 4} \|\Delta_j u\|_{L^\infty} \left(\|\nabla S_{j-1} \phi\|_{L^p} + \|\nabla m(D) S_{j-1} \phi\|_{L^p} \right) \\ & \leq C \|\nabla u\|_{L^\infty} \sum_{j \in \mathbb{N}, |j-q| \leq 4} 2^{-j(\epsilon+1)} \left(\sum_{-1 \leq j' \leq j-1} 2^{j'} \|\Delta_{j'} \phi\|_{L^p} \right) \\ & \leq C \|\nabla u\|_{L^\infty} \sum_{|j-q| \leq 4} \sum_{-1 \leq j' \leq j-1} 2^{(j'-j)(1+\epsilon)} 2^{-j'\epsilon} \|\Delta_{j'} \phi\|_{L^p}, \end{aligned}$$

which leads to the desired inequality

$$\|\mathcal{II}\|_{B_{p,r}^{-\epsilon}} \leq C \|\nabla u\|_{L^\infty} \|\phi\|_{B_{p,r}^{-\epsilon}}.$$

While for Lemma 2.4-(3), it suffices to show (5.29) in [1], that is, for every $\epsilon \in (0, 1)$ and $(p, r) \in [1, \infty]^2$,

$$(0.3) \quad \|\mathcal{II}_1\|_{\tilde{\mathcal{B}}_{p,r,\mathcal{W}}^{-\epsilon,\ell+1}} + \|\mathcal{II}_2\|_{\tilde{\mathcal{B}}_{p,r,\mathcal{W}}^{-\epsilon,\ell+1}} \lesssim \|\nabla u\|_{\mathcal{B}_{\mathcal{W}}^{0,\ell+1}} \|\phi\|_{\mathcal{B}_{p,r,\mathcal{W}}^{-\epsilon,\ell+1}}.$$

In fact, using Lemmas 5.1 and 5.2 in [1], we find that for every $q \geq -1$ and $\lambda \in \{0, 1, \dots, \ell+1\}$,

$$\begin{aligned} & 2^{-q\epsilon} \|\Delta_q (T_{\mathcal{W} \cdot \nabla})^\lambda \mathcal{II}_1\|_{L^p} \\ & \lesssim 2^{-q\epsilon} \sum_{j \in \mathbb{N}, j \sim q} \|(T_{\mathcal{W} \cdot \nabla})^\lambda m(D) \tilde{\psi}(2^{-j} D) (\Delta_j u \cdot \nabla S_{j-1} \phi)\|_{L^p} \\ & \lesssim 2^{-q\epsilon} \sum_{j \in \mathbb{N}, j \sim q} \sum_{\mu=0}^{\lambda} \|(T_{\mathcal{W} \cdot \nabla})^\mu (\Delta_j u \cdot \nabla S_{j-1} \phi)\|_{L^p} \\ & \lesssim 2^{-q\epsilon} \sum_{j \in \mathbb{N}, j \sim q} \sum_{\mu_1 + \mu_2 \leq \lambda} \|(T_{\mathcal{W} \cdot \nabla})^{\mu_1} \Delta_j u\|_{L^\infty} \|(T_{\mathcal{W} \cdot \nabla})^{\mu_2} \nabla S_{j-1} \phi\|_{L^p} \end{aligned}$$

$$\begin{aligned}
& \lesssim \sum_{j \in \mathbb{N}, j \sim q} \sum_{\mu_1 + \mu_2 \leq \lambda} 2^{-j(1+\epsilon)} \\
& \quad \times \left(\sum_{j_1 \sim j} \sum_{\mu_3=0}^{\mu_1} \|(T_{\mathcal{W} \cdot \nabla})^{\mu_3} \Delta_{j_1} \nabla u\|_{L^\infty} \right) \left(\sum_{j' \leq j-1} \|(T_{\mathcal{W} \cdot \nabla})^{\mu_2} \nabla \Delta_{j'} \phi\|_{L^p} \right) \\
& \lesssim c_q \sum_{\mu_1 + \mu_2 \leq \lambda} \left(\sum_{\mu_3=0}^{\mu_1} \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \mu_3}} \right) \\
& \quad \times \left\| \left(\sum_{j' \leq j-1} 2^{(j'-j)(1+\epsilon)} 2^{-j'(1+\epsilon)} \|(T_{\mathcal{W} \cdot \nabla})^{\mu_2} \Delta_{j'} \nabla \phi\|_{L^p} \right) \right\|_{j \in \mathbb{N}}_{\ell^r} \\
& \lesssim c_q \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \ell+1}} \sum_{\mu_2=0}^{\lambda} \|\nabla \phi\|_{\tilde{\mathcal{B}}_{p, r, \mathcal{W}}^{-1-\epsilon, \mu_2}} \\
& \lesssim c_q \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \ell+1}} \|\phi\|_{\tilde{\mathcal{B}}_{p, r, \mathcal{W}}^{-\epsilon, \ell+1}} \lesssim c_q \|\nabla u\|_{\mathcal{B}_{\mathcal{W}}^{0, \ell+1}} \|\phi\|_{\mathcal{B}_{p, r, \mathcal{W}}^{-\epsilon, \ell+1}},
\end{aligned}$$

with $\{c_q\}_{q \geq -1}$ satisfying $\|c_q\|_{\ell^r} = 1$, and similarly,

$$\begin{aligned}
& 2^{-q\epsilon} \|\Delta_q (T_{\mathcal{W} \cdot \nabla})^\lambda \mathcal{I} \mathcal{I}_2\|_{L^p} \\
& \lesssim 2^{-q\epsilon} \sum_{j \in \mathbb{N}, j \sim q} \|(T_{\mathcal{W} \cdot \nabla})^\lambda (\Delta_j u \cdot \nabla m(D) S_{j-1} \phi)\|_{L^p} \\
& \lesssim 2^{-q\epsilon} \sum_{j \in \mathbb{N}, j \sim q} \sum_{\lambda_1 + \lambda_2 \leq \lambda} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_1} \Delta_j u\|_{L^\infty} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_2} \nabla m(D) S_{j-1} \phi\|_{L^p} \\
& \lesssim \sum_{j \sim q} \sum_{\lambda_1 + \lambda_2 \leq \lambda} 2^{-j(1+\epsilon)} \\
& \quad \times \left(\sum_{j_1 \sim j} \sum_{\lambda_3=0}^{\lambda_1} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_3} \Delta_{j_1} \nabla u\|_{L^\infty} \right) \left(\sum_{j'=-1}^{j-1} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_2} \nabla m(D) \Delta_{j'} \phi\|_{L^p} \right) \\
& \lesssim \sum_{j \sim q} \sum_{\lambda_1 + \lambda_2 \leq \lambda} 2^{-j(1+\epsilon)} \left(\sum_{\lambda_3=0}^{\lambda_1} \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \lambda_3}} \right) \\
& \quad \times \left(\|(T_{\mathcal{W} \cdot \nabla})^{\lambda_2} \nabla m(D) \Delta_{-1} \phi\|_{L^p} + \sum_{j'=0}^{j-1} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_2} \nabla m(D) \Delta_{j'} \phi\|_{L^p} \right) \\
& \lesssim \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \ell+1}} \sum_{j \sim q} \sum_{\lambda_2=0}^{\lambda} 2^{-j(1+\epsilon)} \left(\|\nabla m(D) \Delta_{-1} \phi\|_{L^p} + \sum_{j'=0}^{j-1} \sum_{\lambda_4=0}^{\lambda_2} 2^{j'} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_4} \Delta_{j'} \phi\|_{L^p} \right) \\
& \lesssim c_q \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \ell+1}} \\
& \quad \times \left(\|\Delta_{-1} \phi\|_{L^p} + \sum_{\lambda_4=0}^{\lambda} \left\| \left(\sum_{0 \leq j' \leq j-1} 2^{(j'-j)(1+\epsilon)} 2^{-j'\epsilon} \|(T_{\mathcal{W} \cdot \nabla})^{\lambda_4} \Delta_{j'} \phi\|_{L^p} \right) \right\|_{j \in \mathbb{N}}_{\ell^r} \right) \\
& \lesssim c_q \|\nabla u\|_{\tilde{\mathcal{B}}_{\mathcal{W}}^{0, \ell+1}} \|\phi\|_{\tilde{\mathcal{B}}_{p, r, \mathcal{W}}^{-\epsilon, \ell+1}} \lesssim c_q \|\nabla u\|_{\mathcal{B}_{\mathcal{W}}^{0, \ell+1}} \|\phi\|_{\mathcal{B}_{p, r, \mathcal{W}}^{-\epsilon, \ell+1}},
\end{aligned}$$

then the desired inequality (0.3) follows immediately.

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